

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (Backlog) Examination (CL)

November-December 2012

CL204 Engineering Geology

Date: 30.11.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Write a short note on different branches and allied science branches of Geology. [05]
(b) Describe cleavage and fracture of minerals. [02]
- Q - 2 (a) With the help of neat diagrams, describe concordant and discordant igneous forms. [07]
(b) Write a short note on Bowens Reaction Series. [04]
(c) Define rock forming minerals and ore minerals with suitable example? [03]

OR

- Q - 2 (a) Describe the physical properties of mineral depends on light. [06]
(b) Write a short note on common silicate structures. [06]
(c) Define volcanic igneous rocks with a suitable example. [02]
- Q - 3 (a) Describe the geological features associated with the action of wind. [08]
(b) Write a short note on earthquake waves. [06]

OR

- Q - 3 (a) Write a short note on the internal structure of earth. [06]
(b) Describe the different methods by which we can get the information of internal structure of earth. [06]
(c) Define seismogram. [02]

SECTION – II

- Q - 4 (a) Write note on internal factors controlling mass movements. How the problem of mass movement in hilly region can be averted? [07]
- Q - 5 (a) Define fold. Describe types of fold. Add note on causative factors of folding. [07]
(b) Describe unconformity and its types. Write their geological importance. [07]

OR

- Q - 5 (a) Define fault. Draw the sketch of an ideal fault, reverse fault, normal fault and define their parts. How faults are recognized in the field? [07]

- (b) What are joints? State the classification of joints based on genesis. Write its geological importance. [07]
- Q - 6 (a) Define geological maps and topographical maps. Which are the components / units seen in geological maps? Write its engineering importance. [07]
- (b) Give classification of dams. Describe various forces that act on dams influenced by geological factors. [07]

OR

- Q - 6 (a) Describe geological investigation required for the site selection of the tunnel alignment. How does water level control the alignments of the tunnel? [07]
- (b) Write the objectives of Engineering Geology. Explain in detail the influence of fold and fault on various civil engineering structures. [07]
